

Exercise 03: Relational algebra, DML and DDL

Task 1*: Creating tables using the SQL Data Definition Language

In the first exercise, you thought about views that you need in order to present data for both the conceptual and external layers. Now you will be creating the conceptual layer tables in MS SQL.

- a) In your database, create a `Students` table with the following attributes:
Name of type variable string with a maximum of 30 characters.
Matriculation of type integer with a maximum of 4 characters. This attribute is the primary key and cannot be negative.
- b) In your database, create a `Lecturers` table with the following attributes:
Name of type variable string with a maximum of 30 characters. This attribute is the primary key.
Office of type variable string with a maximum of 30 characters. This attribute should always be specified.
Tel of type variable string with a maximum of 30 characters. This attribute can also be left empty.
- c) In your database, create an `Events` table with the following attributes:
Name of type variable string with a maximum of 30 characters. This attribute is part of the primary key.
Semester of type constant string with a maximum of 4 characters. This attribute is part of the primary key.
Room of type variable string with a maximum of 8 characters.
Lecturer is a foreign key for the name from `Lecturers`. When the name of the lecturer changes in table `Lecturers`, the name shall also change here. When the lecturer is removed from `Lecturers`, the event has no lecturer.
- d) In your database, create the `Student_in_Event` table with the following attributes and prevent a student from registering for the same event more than once.
Student is a foreign key for the matriculation number from `Students`. When the matriculation number in `Students` changes, the changes shall propagate to here and if the student is removed, all its grades shall also be deleted.
Event is a foreign key for the name from `Events`.
Semester is a foreign key for the semesters from `Events`. If an events name or semester changes, the changes shall propagate. If an event is removed, all grades shall also disappear.
Grade is a decimal number with a total of 2 digits and one of the digits after the decimal point. The grade should not be less than 1 and not greater than 5.
ID is an automatically integrated primary key starting from 10 only allowing even numbers.

Task 2*: Insert, modify and delete tuples using the SQL Data Manipulation Language

Now make the following changes to the data in the previously created database:

- a) Professor Klaus is based in room C201 and can be reached at telephone number 123. In the winter semester 2017 (ws17) and the summer semester 2018 (ss18), he holds the dance gymnastics event in room D111. In the winter semester 2018, he holds the sack race event. The room is still to be confirmed.
- b) The hang-gliding and beach volleyball events are offered by Professor Maria in the summer semester 2017 as well as the summer semester 2018. She is based in room D120. Her events always take place at the beach.
- c) The students Eva, Luise, Daniel and Dominik have the matriculation numbers 3333, 3334, 3335 and 3336.
- d) Eva, Luise and Daniel take beach volleyball in the summer semester 2018
- e) Dominik and Eva take hang-gliding in the summer semester 2017
- f) Luise and Daniel take dance gymnastics in the winter semester 2017 and beach volleyball in the summer semester 2017
- g) It's not possible to fail beach volleyball, so all students in this course should get a grade of at least 4.0.
- h) Professor Maria has a new office in room D22.
- i) (additional task) Eva has moved and de-registers herself from the university. De-register Eva from all events.

Task 3: Subsequent modification of relational schemas using SQL DDL

After you have been using your database for a while, you realise that you would like to make some changes to your conceptual schema. Use `alter table` to make the following changes:

- a) Add a `Date_of_birth` field of the type `date` to the Students table.
- b) Insert into the table that Eva was born on 1 March 1990, Luise a month later, Daniel a month after Luise and Dominik a month after Daniel.
- c) In a second step, you think that you also want to always force a date of birth to be specified. Change the table accordingly. What must you consider when doing so?

Task : Generative AI

Feed Task 1 into generative AI and compare your solution to the generated one.